

EC5202 – Digital Systems Design with HDL

2:1 Multiplexer Design and Verification

1.Verilog Module

```
mux.v •  
mux.v  
1 // 2:1 Multiplexer: selects one of two inputs  
2 module mux (  
3     input wire a,      // First input  
4     input wire b,      // Second input  
5     input wire sel,    // Select signal  
6     output wire y      // Output  
7 );  
8  
9     // If sel = 0, y = a. If sel = 1, y = b.  
10    assign y = sel ? b : a;  
11  
12 endmodule
```

2. Test Bench Code

```
mux.v • mux_tb.v •
mux_tb.v
1  `timescale 1ns/1ps
2  module mux_tb;
3      reg a, b, sel;
4      wire y;
5
6      mux uut (.a(a), .b(b), .sel(sel), .y(y));
7
8      initial begin
9          $dumpfile("mux.vcd");
10         $dumpvars(0, mux_tb);
11         $display("Time sel a b y");
12         $monitor("%4t %b %b %b %b", $time, sel, a, b, y);
13
14         // Test sel = 0: y should equal a
15         sel=0; a=0; b=0;
16         #10 a=0; b=1;
17         #10 a=1; b=0;
18         #10 a=1; b=1;
19
20         // Test sel = 1: y should equal b
21         #10 sel=1; a=0; b=0;
22         #10 a=0; b=1;
23         #10 a=1; b=0;
24         #10 a=1; b=1;
25         #10 $finish;
26     end
27 endmodule
```

3. Output in cmd

```
Problems  Output  Debug Console  Terminal  Ports

• PS D:\VerilogPractice> $env:Path += ";D:\iverilog\bin"
• PS D:\VerilogPractice> iverilog -s mux_tb -o mux_sim.vvp mux.v mux_tb.v
• PS D:\VerilogPractice> vvp mux_sim.vvp
VCD info: dumpfile mux.vcd opened for output.
Time sel a b y
  0    0  0  0  0
10000   0  0  1  0
20000   0  1  0  1
30000   0  1  1  1
40000   1  0  0  0
50000   1  0  1  1
60000   1  1  0  0
70000   1  1  1  1
mux_tb.v:41: $finish called at 80000 (1ps)
• PS D:\VerilogPractice> 
```

4. Waveform

